

SFA1200AJ

Laser Rangefinder Module



Product Manual

Version 202606V1

Disclaimer

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Safety Instructions

For the safe use of the ranging module, please carefully follow the safety instructions provided in this manual. Do not attempt to disassemble, modify, or repair any part of the ranging module. Do not tamper with or attempt to alter the performance of the ranging module.

Intended Use

- Take precautions against electrostatic discharge (ESD). Do not touch any electronic components or parts of the ranging module without proper protective measures.
- Keep this manual readily accessible to all personnel operating the ranging module.
- Operate the ranging module only within the specified voltage and power range.
- Operate the ranging module in a clean, dry, and ESD-protected environment.
- Do not touch the optical glass lens with your fingers.
- Avoid exposing the ranging module to environments with drastic temperature fluctuations.

Improper Use

- Using the ranging module with the host system before verifying that all functions are operating properly;
- Operating the ranging module without first reading and understanding this manual and all safety instructions;
- Disassembling, modifying, or altering the ranging module without authorization from SunFlaser Technology (Chengdu) Co., Ltd.;
- Sending operating commands other than those specified in this manual;
- Using or testing the ranging module in non-standard, abnormal, or unsuitable environments;

- › Operating the ranging module in a manner inconsistent with the instructions provided in this manual;
- › Continuing to operate the ranging module after improper handling or misuse;
- › Using a ranging module with obvious faults, defects, or damage;
- › Performing test operations within the blind zone of the ranging module.

Improper use may result in the following risks:

- › Eye injury
- › Incorrect results
- › Measurement errors
- › Property damage
- › Equipment malfunction

Operator Responsibilities

- › Be familiar with the basic knowledge required for the correct operation of the ranging module;
- › Understand and comply with the operation and maintenance requirements of the ranging module;
- › Prevent damage to the ranging module during storage and transportation;
- › Keep the ranging module clean and well maintained;
- › Ensure that the ranging module is stored and operated in a safe environment;
- › Do not expose the ranging module to strong mechanical shock or impact;
- › Do not expose the ranging module to environments with drastic temperature fluctuations.

Avoid Measurement Errors

- › Pay attention to all factors that may affect the performance of the ranging module;

- After the Laser Rangefinder Module has been subjected to improper handling (such as vibration, impact, or dropping), or before performing critical measurement tasks, it is recommended to verify the performance specifications of the ranging module.

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01

Product Overview

The SFA1200AJ Laser Rangefinder Module is a new lightweight and compact ranging module operating at a wavelength of 905 nm. The product provides a maximum ranging distance of ≥ 1200 m and features a UART-TTL interface. It is supplied with dedicated test software for convenient secondary development by users.

With its compact size, light weight, and reliable performance, the module is suitable for applications in aviation, communications, geology, law enforcement, and outdoor sports.

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Key Features

> Long-Range Distance Measurement

The maximum ranging distance reaches up to 1200 m, while the minimum ranging distance supports short-range measurement, making it suitable for both long-range surveying and short-range detection applications.

> High-Precision Measurement

Measurement accuracy: $\pm (0.7 + 0.004 \times d)$ m when $d > 100$ m. The measurement frequency is stable, with an error rate $\leq 1\%$.

> Flexible Measurement Modes

Supports single ranging and continuous ranging modes, with customizable measurement intervals to meet different testing requirements.

➤ Low Power Consumption Control

Features a low-power mode with standby power consumption ≤ 1 mW. During normal operation, the power state can be switched through the enable pin.

➤ Standard Serial Communication

Adopts a UART-TTL (3.3 V) interface with a default baud rate of 115200 bps. Multiple baud rate configurations are supported, facilitating embedded system integration and development.

➤ Comprehensive Development Support

Provides dedicated test software and a complete communication protocol, supporting power-on self-test and LD continuous-on mode debugging, which helps reduce development complexity.

➤ Safety and Compliance

The laser classification complies with Class 1 (IEC 60825-1), posing no risk of eye injury and making it suitable for integration into a wide range of civilian applications.

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Key Specifications

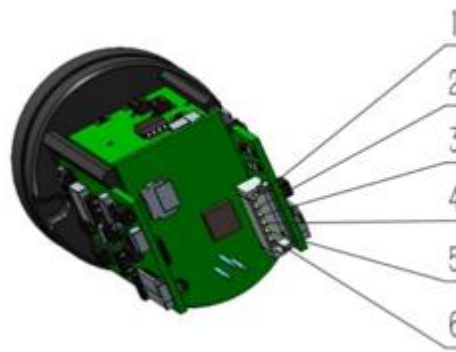
Item		Parameter
Electrical Specifications	Maximum Ranging Distance (Visibility ≥ 6 km)	≥ 1200 m @ 70% reflectivity, building target
	Minimum Ranging Distance	5 m @ 90% reflectivity, white target board
	Ranging Accuracy (Building Target)	$d \leq 100$ m: ± 0.7 m $d > 100$ m: $\pm(0.7 + 0.004 \times d)$ m
	Measurement Frequency	3 Hz when continuously measuring the same target 2 Hz when switching between different targets
	Measurement Success Rate	$\geq 98\%$
	False Alarm Rate	$\leq 1\%$
	Baud Rate	115200 bps (Factory Default) / 9600 bps
	Interface Type	UART(TTL_3.3 V)
	Power Supply Voltage	DC 3.3 $\pm$ 0.1 V
	Standby Power Consumption	≤ 400 mW (POWER_ON pulled low)
		≤ 1 mW (POWER_ON pulled high)
	Operating Power Consumption (+3.3 V in)	≤ 1.5 W@25 °C ≤ 1.8 W@70 °C ≤ 1.3 W@-20 °C
	Response Time (System Startup + First Measurement)	Minimum: ≤ 500 ms + 46 ms (short range) Maximum: ≤ 500 ms + 458 ms (long range)
Optical Specifications	Laser Wavelength	905 $\pm$ 10 nm
		Class 1(IEC 60825-1)
	Laser Beam Divergence (Typical)	≤ 10 mrad
	Average Laser Power	8.23 mW
	Receiver Field of View (FOV)	≈ 8 mrad
Physical	Weight	22 $\pm$ 0.5 g

Characteristics	Dimensions	Ø23 mm × 40 mm
	Shock Resistance	1000 g/ms (10 times/s along the optical axis)
	Vibration Resistance	5–50–5 Hz, 1 octave/min, 2.5 g
Environmental Adaptability	Operating Temperature Range	-20 to +60 °C
	Storage Temperature Range	-25 to +75 °C
	Reliability	MTBF ≥ 1500 h
	Ingress Protection Rating (IP Rating)	IP67

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Interface and Connection

Communication Interface: UART (TTL 3.3 V), default baud rate: 115200 bps.

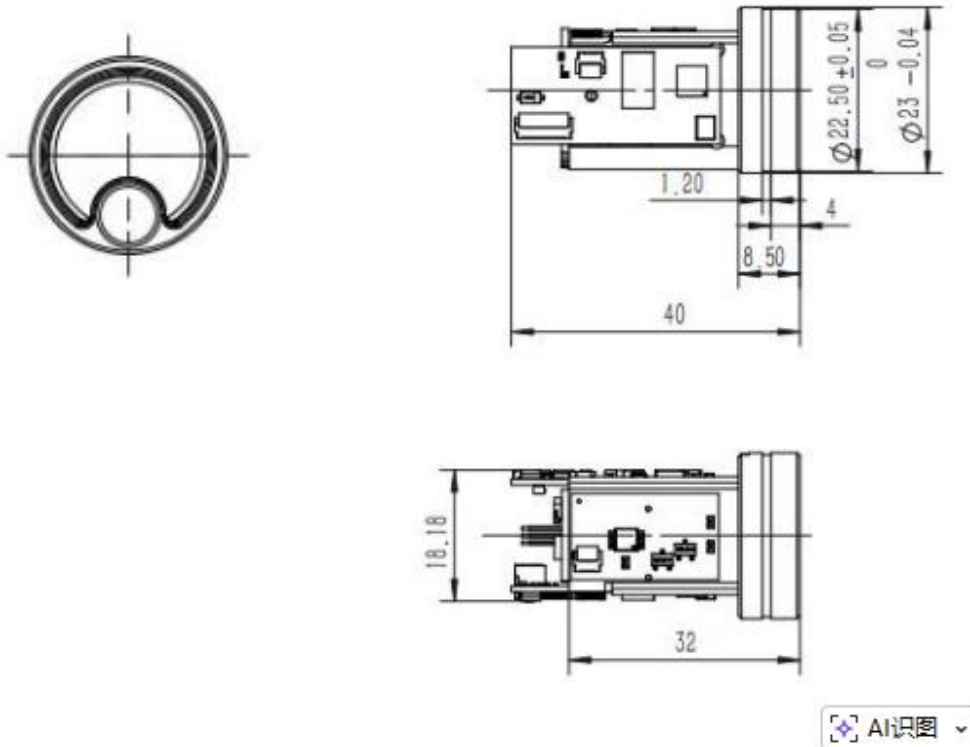


Interface Definition:

Pin Assignment	Function Definition	Description
Pin 1	GND	Ground
Pin 2	Power	3.3 V DC Power Supply
Pin 3	NC	No Connection (NC)
Pin 4	TTL_TXD	Serial Transmit (TX), TTL 3.3 V Level
Pin 5	TTL_RXD	Serial Receive (RX), TTL 3.3 V Level
Pin 6	Enable Pin	Power-On with Low Level

Low-Power Mode:

- In this mode, the device consumes very little power. The MCU is powered down and does not respond to any commands.
- To perform a measurement, pull the enable pin low to enter normal operating mode.
- After completing the measurement, pull the enable pin high to return the device to low-power mode, with power consumption below 1 mW.
- Mechanical Interface and Center of Gravity:



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Communication Protocol and Commands

For details of the following contents, please refer to Appendix 1:

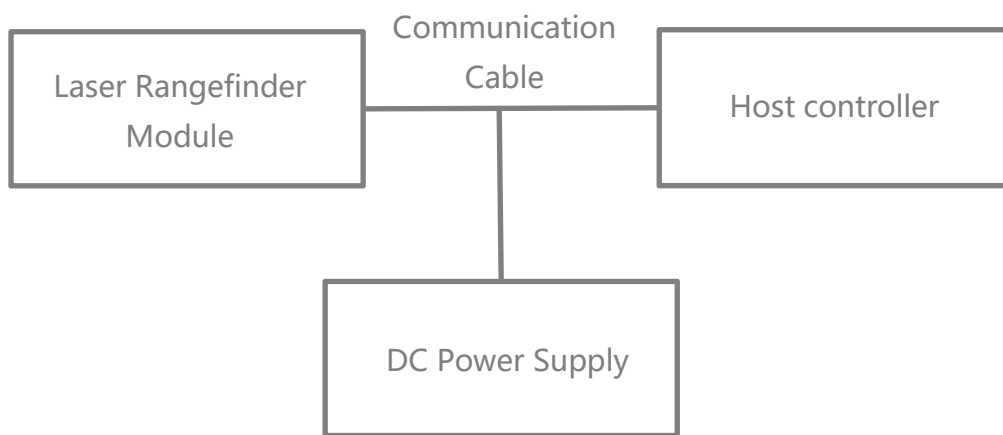
- Testing Guide
- Protocol Format

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Operating Instructions

Power-On Procedure

- Before powering on, connect the Laser Rangefinder Module, communication cable, DC power supply, and Host Controller according to the electrical interface requirements and as shown in the figure below.



- The module powers on automatically after the power supply is connected.

Power-Off Procedure

- Before powering off, ensure that all operating processes and tasks have been completed and that the program has been exited properly.
- After confirmation, disconnect the power supply to power off the module normally.

Ranging Mode Operation

- Send the “Single Ranging” command to the Laser Rangefinder Module. The module performs a single ranging operation and returns the ranging status and distance value.
- Send the “Continuous Ranging” command to the Laser Rangefinder Module. The module performs continuous ranging and returns the ranging status and distance value continuously.
- Send the “Stop Ranging” command to terminate the ranging operation.

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Precautions for Use

- Do not stare directly into the laser emission aperture to avoid eye injury (compliant with Class 1 standards).
- Operating temperature range: -20 ° C to +60 ° C. Avoid exposure to high temperature, humidity, rain, snow, or dusty environments.
- Do not disassemble the module without authorization, as this may void the warranty or cause device damage.
- This product is mainly intended for integration into civilian UAVs, outdoor observation, engineering surveying, and related equipment. Use in any military applications is prohibited.
- The spot adjustment/debug mode is intended for development personnel only. Do not aim the laser at human eyes while this mode is enabled to avoid eye injury.
- Factors affecting ranging capability, measurement response speed, and speed measurement accuracy include:
 - Target Reflectivity: Generally, the higher the target's reflectivity, the better the ranging capability and the faster the measurement response. For example, a medium-reflectivity target can be measured up to 1200 m, a high-reflectivity target can reach at least 1500 m, and a low-reflectivity target may only reach 600 m. Targets that are difficult to produce diffuse reflection, such as water surfaces, may be unmeasurable.
 - Target Shape: When the reflective surface area of the target is too small or uneven, both the ranging capability and the measurement response speed will decrease accordingly.

- **Measurement Angle:** The ranging capability and measurement response speed are optimal when the laser beam strikes the target's reflective surface perpendicularly. Conversely, both the ranging capability and response speed decrease at oblique angles. Under extreme measurement angles, the module may not achieve the ranging performance and response speed specified in this manual.
 - **Measurement Environment:** Factors affecting ranging capability and measurement response speed also include sunlight intensity, the concentration of water vapor and suspended particles in the air, and the angle relative to sunlight. For example, conditions such as rain, fog, snow, or haze can reduce the effective measuring distance.
- The ranging distance of this series of rangefinder binoculars is defined under the following conditions:
- The measurement target has medium reflectivity, such as a building wall.
 - The reflective surface of the measurement target is perpendicular to the direction of laser emission.
 - The measurement is conducted in clear weather conditions without direct sunlight exposure.

Recommendation: When measuring long-distance targets, it is recommended to mount the module on a tripod to minimize vibration during operation, thereby achieving more stable and accurate measurement results.

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Optical Window Requirements

> Material Recommendations

The optical window material is recommended to be Chengdu Guangming Optical Glass H-K9L or fused silica. H-K9L and fused silica are commonly used colorless optical glasses suitable for laser wavelengths from 300 nm to 2100 nm. They offer high cost-effectiveness and excellent physical properties.

> Manufacturing Recommendations

The wedge angle tolerance of the optical window should be as small as possible. It is recommended that the wedge angle tolerance be $\leq 3'$ (tolerance grade ≤ 7). The optical surface of the window should be as smooth as possible, with a recommended arithmetic average roughness (Ra) of 0.012 μm .

> Coating Recommendations:

For 905 nm laser rangefinder modules, it is recommended to apply an anti-reflection (AR) coating optimized for the 855 nm–955 nm wavelength range, with a transmission rate of $\geq 99.5\%$.

Depending on the specific application environment, additional protective coatings such as hydrophobic coatings or hard coatings may be applied to the outer surface of the optical window. Other performance requirements should comply with GJB 2485-95, with a transmission rate of $\geq 98\%$.

> Optical Window Shape and Usage Recommendations:

The effective aperture of the optical window varies depending on different products. Its overall dimensions should satisfy the following requirements: the difference between the optical window effective aperture and the outer diameter should be ≥ 2 mm; and the difference between the rangefinder antenna outer diameter and the projected size of the optical window effective aperture should be ≥ 1.5 mm, as shown in the schematic diagram below.

Since the optical window has a certain level of laser absorption, it is recommended that the thickness of the optical window be controlled within 2–4 mm according to the overall size.

Due to the high transmittance of the optical window, it is recommended that the emission optical axis be aligned parallel to the normal of the optical window surface. The positioning relationship between the optical window and the dual optical barrels is shown in the schematic diagram below.

Meanwhile, the air gap between the optical window and the rangefinder should be less than 0.5 mm. The figure below illustrates two installation configurations of the optical window.

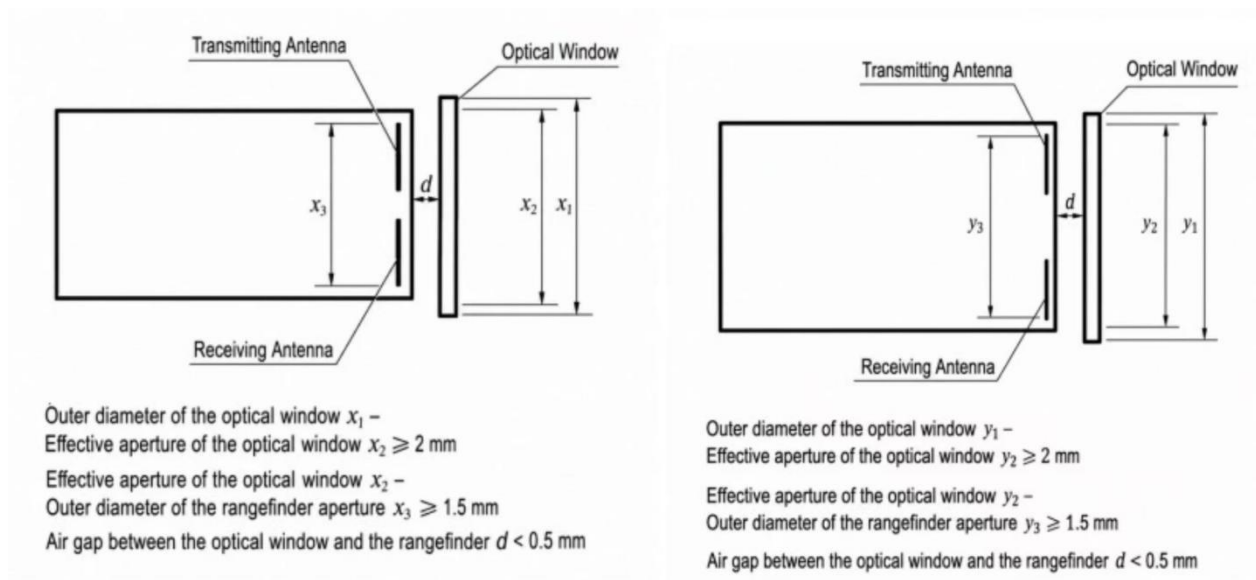


Figure 1 : Optical Window Dimensions and Placement Diagram

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Safety Precautions

- › To ensure the safety of both operators and test subjects, the following safety measures have been incorporated into the design of the SFB2000A Laser Rangefinder Module;
- › Safety design and analysis are conducted in accordance with GJB 900A-2012, General Requirements for Equipment Safety;
- › Non-flammable materials are used, and all mechanical and electrical interfaces are securely connected;
- › Appropriate design measures are implemented to prevent moisture accumulation that could result in short circuits;
- › The module operates below the safety voltage limit for the human body.

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Maintenance Instructions

Packing List

The standard items supplied with the product are listed in the table below. Optional accessories may not be included in this list and shall be subject to the actual product delivered.

No.	Item	Quantity	Remarks
1	Laser Rangefinder Module	1 unit	-
2	Certificate of Conformity	1 copy	-
3	Packaging Box	1 pc	-
4	Power Cable	1 pc	-

Module Cleaning

Cleaning of Optical Components

- Dust particles should be removed using an air blower;
- Fingerprints should first be cleaned with absorbent cotton lightly moistened with a small amount of alcohol-ether mixture, and then wiped with a clean lens cloth.

Cleaning of Mechanical and Electronic Components

- With the power disconnected, gently wipe mechanical and electronic components with alcohol and allow them to dry naturally before use;
- Keep the ranging module, connectors, and cables away from moisture and contaminants whenever possible;
- Ensure that the equipment is completely dry before packaging.

Inspection and Maintenance

Routine Maintenance

When the product is used for the first time, or after replacement of related modules or components, both visual inspection and power-on inspection shall be performed. For products under normal operation, only a power-on inspection is required before use.

- Visual Inspection Procedure
 - Check whether the product appearance is normal;

- Check whether the cable connections are correct and securely connected.
- Power-On Inspection Procedure
 - Perform the power-on procedure and verify that the standby current is correct;
 - Start the self-test function of the product;
 - After the inspection is completed, perform the power-off procedure.

Periodic Maintenance

Under normal operating conditions, the Laser Rangefinder Module does not require routine maintenance. If the module has been stored in a dust-free environment for more than one year, maintenance should be performed, including general inspection and power-on inspection.

- General Inspection (Performed with Power Off)
 - All markings and serial numbers on the product and test cable connectors shall be correct and clearly legible;
 - All screws on the panel shall be securely fastened;
 - The optical glass of the product shall be free from visible defects that may affect normal operation, such as spots, pitting, water stains, fungus, fingerprints, dust particles, or cracks.
- Comprehensive Power-On Inspection and Maintenance
 - Connect the product power supply in sequence;
 - Perform the power-on procedure and verify that the standby current is correct;
 - Start the product self-test and complete the self-test procedure;
 - Perform the power-off procedure after completion of the inspection.

Troubleshooting

The following list only includes possible fault phenomena not caused by the product itself. If any other technical problems occur during operation, please contact our company for technical support.

Issue	Possible Cause / Solution
Measured distance is abnormally shorter than the actual distance	Testing in heavy haze, fog, or high-humidity environments may cause the module to detect airborne water droplets or particles instead of the actual target. It is recommended to perform testing in clear environmental conditions.
Ranging performance decreases, or the theoretical target distance cannot be reached	① Very low target reflectivity can reduce the returned laser energy and affect ranging performance. When measuring low-reflectivity targets such as lake surfaces, wet roads, or glass, the actual performance may not achieve the theoretical range.
	② If the ranging module is not aimed perpendicular to the target surface and instead measures at an angle, the returned laser energy will attenuate under oblique measurement conditions, resulting in reduced ranging performance.
	③ Since the laser wavelength overlaps with part of the solar spectrum, direct exposure of the ranging module to sunlight may reduce ranging performance.
No response to commands when communicating with a PC using the supplied communication cable set	Check whether the communication cable is damaged or broken (the 6-pin communication cable is relatively thin and can break easily).
No response to commands when using a custom communication cable	Check whether the external wiring sequence is correct, and verify that the communication cable has no short circuit, open circuit, or poor contact. Also confirm that the UART TX and RX lines are not reversed.

Packaging and Transportation Requirements

> Packaging

After unpacking, if the product needs to be placed back into storage, it shall be repackaged using the original packaging whenever possible.

If the product needs to be returned to the manufacturer, the original packaging is recommended. If alternative packaging is used, it shall not cause any degradation in product performance or physical damage.

> Transportation

Repackaged products may be transported by automobile, railway, aircraft, or ship. During transportation, the package shall be securely fixed to the transport vehicle to avoid impact, improper handling, and exposure to rain or snow.

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Technical Support and Service

If a product malfunction occurs and factory return is required for fault analysis, troubleshooting, or repair, SunFlaser Technology (Chengdu) Co., Ltd. provides a one-year warranty and lifetime technical support from the date of product delivery.

During the warranty period, product failures caused by manufacturing or quality issues will be repaired or replaced free of charge by SunFlaser. For product damage caused by improper operation or other user-related factors, repair and replacement costs will be charged according to the actual situation.

This product is a precision optical instrument. Please handle and operate it carefully during use. If you have any additional questions regarding operation or maintenance, please contact our after-sales support personnel at any time.

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Contact Information

Website: www.sunflaser.com

Tel: +86 186 2827 7862 (Sales Manager: Mr. Cao)

Address: No. 58, Zhenxing West 1st Road, Building 1, Room 201, Jinniu District, Chengdu, Sichuan, China

Appendix 1:

> Testing Guide

Operating Procedure

Step 1: Connect the data cable to the ranging module. The module will be powered on automatically, and measurement data can be output simultaneously.

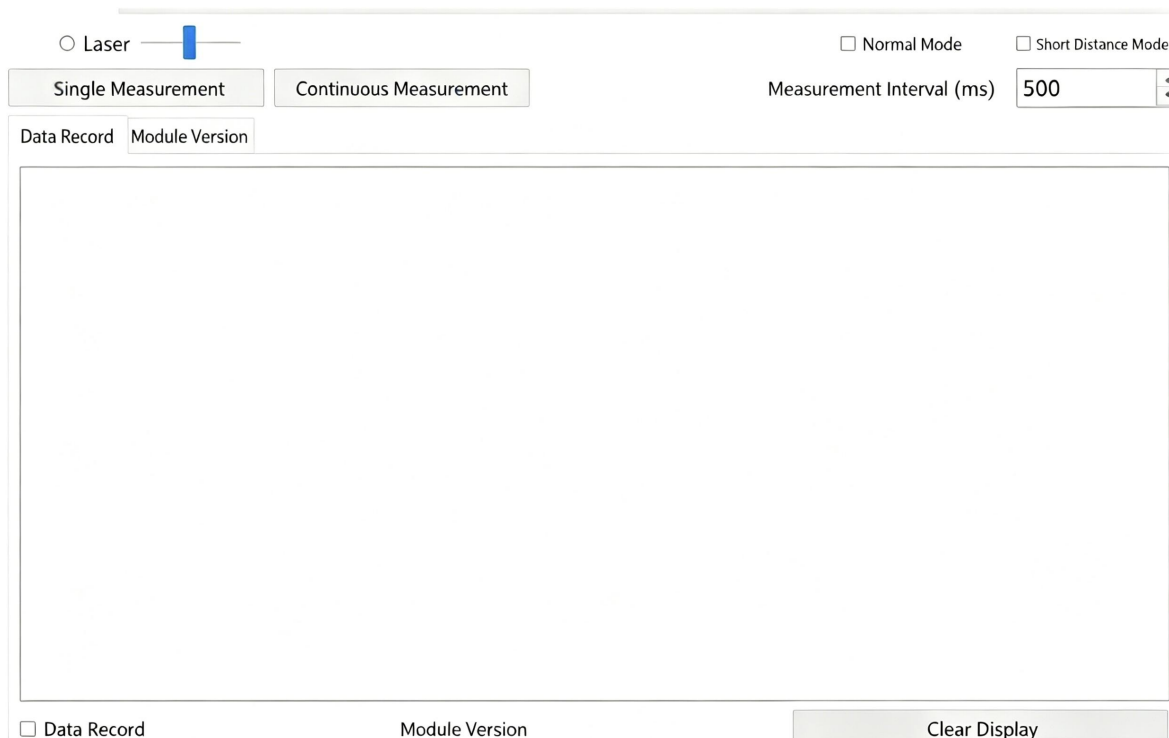
Note 1: Use the original factory-standard cable and strictly maintain the supply voltage within 3.3 ± 0.1 V.

Note 2: The enable pin must be pulled low (connected to GND).

Step 2: Install the software, and use a CH340 or CP2102 module to convert the serial port signal to USB for connection to a computer.

Note 1: Connect the module TXD to the RXD of the TTL-to-USB converter, and connect the module RXD to the TXD of the TTL-to-USB converter.

Step 3: After the software is installed, open the display interface.



The screenshot shows the SunFlaser software interface. At the top, there is a 'Laser' toggle switch (currently off) and two checkboxes: 'Normal Mode' and 'Short Distance Mode'. Below these are two buttons: 'Single Measurement' and 'Continuous Measurement'. To the right of these buttons is a 'Measurement Interval (ms)' field set to '500'. Below the buttons are two tabs: 'Data Record' and 'Module Version'. The main area of the interface is a large empty rectangle. At the bottom, there are three elements: a 'Data Record' checkbox, a 'Module Version' label, and a 'Clear Display' button.

Step 4: Debugging and Testing

- Select the serial port: Choose the corresponding COM port in the software according to the computer's serial port number.
- Baud rate setting: After opening the software interface, configure the baud rate. Available options include 9600, 115200, etc. (see Communication Protocol).

➤ Protocol Format

Communication Method: Serial communication

Baud Rate: 115200 (default)

Data Bits: 8 bits

Frame Length: 8 bytes

Data Protocol								
	Header(H)	Header(L)	Function Code	D1	D2	D3	D4	Checksum
Transmission	55	AA						SUM (Function Code + DATA1 + ... + DATA4)
Response	55	AA						SUM (Header H + Header L + ... + DATA4)

Measurement Command									
Single Ranging	Transmit	55	AA	88	FF	FF	FF	FF	SUM[3: 7]
		55 AA 88 FF FF FF FF 84							
	Response	55	AA	88	STA	FF	DIS_H	DIS_L	SUM[1: 7]
		STA = Others: Measurement failed; STA = 1: Measurement successful. DIS_H: High byte of the measurement result; DIS_L: Low byte of the measurement result. The returned data is transmitted in hexadecimal format. All measurement results are output as the actual value multiplied by 10.							
Continuous Ranging	Transmit	55	AA	89	FF	FF	FF	FF	SUM[3: 7]
		55 AA 89 FF FF FF FF 85							
	Response	55	AA	89	STA	FF	DIS_H	DIS_L	SUM[1: 7]
		STA = Others: Measurement failed; STA = 1: Measurement successful. DIS_H: High byte of the measurement result; DIS_L: Low byte of the measurement result. The returned data is transmitted in hexadecimal format. All returned measurement values are scaled by a factor of 10 relative to the actual value.							
Stop Ranging	Transmit	55	AA	8E	FF	FF	FF	FF	SUM[3: 7]
		55 AA 8E FF FF FF FF 8A							
	Response	55	AA	8E	STA	FF	FF	FF	SUM[1: 7]
		STA = 0: Failed to stop continuous ranging. STA = 1: Continuous ranging stopped successfully.							
Angle Measurement	Transmit	55	AA	8A	FF	FF	FF	FF	SUM[3: 7]
		55 AA 8A FF FF FF FF 86							
	Response	55	AA	8A	STA	FF	ANG_H	ANG_L	SUM[1: 7]
		STA = 0: Measurement failed; STA = 1: Measurement successful. ANG_H: High byte of the angle measurement result; ANG_L: Low byte of the angle measurement result. The returned data is transmitted in hexadecimal format. All returned measurement values are scaled by a factor of 10 relative to the actual value. This function is only available for modules equipped with an angle sensor.							

Power-On Self-Test									
Self-Test Information	Response	55	AA	80	STA	00	00	ErrCode	SUM[1: 7]
		STA = 0: Power-on initialization failed; ErrCode indicates the error code.							
		STA = 1: Power-on initialization successful.							

System Configuration									
Baud Rate	Transmit	55	AA	TYPE	FF	FF	FF	FF	SUM[3: 7]
		TYPE = 01: Set baud rate to 9600 bps							
		TYPE = 02: Set baud rate to 14400 bps							
		TYPE = 03: Set baud rate to 19200 bps							
		TYPE = 04: Set baud rate to 38400 bps							
		TYPE = 05: Set baud rate to 56000 bps							
		TYPE = 06: Set baud rate to 57600 bps							
		TYPE = 07: Set baud rate to 115200 bps							
		TYPE = 08: Set baud rate to 128000 bps							
		TYPE = 09: Set baud rate to 230400 bps							
		The baud rate setting does not take effect immediately. The new baud rate will become effective only after the device is restarted.							
	Response	55	AA	TYPE	STA	FF	FF	FF	SUM[1: 7]
		STA = 0: Configuration failed.							
		STA = 1: Configuration successful.							
LD Continuous-On Mode	Transmit	55	AA	86	FF	FF	FF	STA	SUM[3: 7]
		STA = 0: LD continuous-on mode disabled.							
		STA = 1: LD continuous-on mode enabled.							
		55 AA 86 FF FF FF 01 84 — Enable LD Continuous-On Mode							
		55 AA 86 FF FF FF 00 83 — Disable LD Continuous-On Mode							
	Response	55							
		STA = 0: LD continuous-on mode is enabled.							
		STA = 1: LD continuous-on mode is disabled.							
		STA = FF: Operation failed.							
		Note: Do not enable LD continuous-on mode for extended periods of time.							

- The checksum content for transmitted and received data is different; please distinguish them carefully.
- The checksum is defined as the sum of the bytes to be verified, with only the lowest 8 bits retained.
- All data is transmitted and received in hexadecimal format.